State Methodology (10/22)

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This appendix explains how I adapt the Budget Lab's national ECEC model to local markets. I use the national methodology's definitions and notation. The baseline is 2019. Each run covers children ages 0–4 in one of the 50 states, DC, or New York City (NYC). NYC is the five boroughs; it is also included in New York State, so I exclude its separate run when adding up national totals.

I keep the nationally estimated forests, utility function, β, ρ, production technology, and care-worker labor-supply elasticities. I change local care-hour targets, wages, staffing requirements, non-labor costs, and parents' employment paths. Each geography is a separate market, without movement of families or workers across its boundary.

1 Income ranks and national calibration

1. **Put incomes on a comparable purchasing-power basis.** I divide 2019 household income by the 2019 BEA all-items regional price parity (RPP), expressed relative to the US average of 100:

$$\text{income used for ranking} = \frac{\text{dollar income}}{\text{all-items RPP}/100}.$$

   An income of $55,000 in a place with RPP 110 becomes $50,000 for ranking. I make this adjustment because the same dollar income buys less in a more expensive place. The rank is an input to the prediction models. The household budget still uses dollar income, with care payments deducted separately.

2. **Assign the available geographic price index.** ACS households use their state's 2019 BEA all-items RPP. NSECE identifies only four Census regions, so each NSECE household uses its region's average state RPP. I weight that average by 2019 ACS children under five living in households, using child person weights. NYC uses the New York–Newark–Jersey City metropolitan RPP, 114.9; that metro area includes parts of New Jersey and Pennsylvania and is a proxy for NYC prices.

3. **Calculate national income ranks within each dataset.** I calculate the 1st through 99th percentile cutoffs of adjusted household income among children under five with positive household income, using child weights. I pool all regions for these cutoffs and calculate them separately in NSECE and ACS. A rank therefore compares a family with families nationally, rather than just families in its own state. I keep the 2019 ranks fixed for the trained forests.

4. **Calibrate the utility parameters nationally.** After this income adjustment, I estimate one β and one ρ, shared across geographies and one-child and two-or-more-child parent units. The 19 October 2026 calibration gives $\beta = 0.0003405$ and $\rho = 0.0262383$.

Aggregate elasticity	Target	Achieved
Employment with respect to paid care cost	−0.15	−0.1500
Work hours with respect to after-tax income	−0.05	−0.0500
Paid care hours with respect to paid care cost	—	−1.5137
Employment with respect to caregiver wages	—	1.5636

I calculate these responses using parent-unit weights. The care-cost experiment raises paid care costs by 1%; the income experiment adds 1% of each family's expected after-tax income as a lump-sum transfer. The first two responses determine the calibration; the other two are model-implied responses.

2 Center price tiers

1. **Define 36 cutoffs.** I use four Census regions, three community density groups, and three child age bands: 0–1, 2–3, and 4–5. States in the same region use the same cutoff for a given density and age. I let cutoffs vary because a given hourly price signals a different position in the local care market across regions, density groups, and child ages. The tiers use price as a proxy for quality.

2. **Measure each center's price and enrollment.** For 2019 NSECE center-based providers, I divide the standard weekly posted rate at each single year of age by 40. Within each two-year age band, I take the simple mean of the available hourly rates. I use positive prices to calculate the medians below. The median weight is the provider's survey weight times full-time-equivalent enrollment in that age band. Full-time enrollment counts as one and part-time enrollment as one-half; I use the mean of the available single-age full-time shares within the band.

3. **Combine regional and density differences.** Let $m_{r,a}$ be the weighted median posted price among high-density centers in region r and age band a. Let $M_{d,a}$ be the weighted median among centers nationally in density group d and age band a. I calculate

$$T_{r,d,a} = m_{r,a} \frac{M_{d,a}}{M_{\text{high},a}}.$$

   I use the national density ratio because the regional rural cells contain only 5–35 surveyed centers per age band, and moderate-density cells contain 28–134 (unweighted counts of centers).

4. **Assign each care arrangement to a tier.** A center uses its own region and density group. A family uses its household's region and density group and its child's age band. I compare posted hourly prices for providers and weekly out-of-pocket payment divided by weekly care hours for households. A price at or above T is high-price. Household payments are net of third-party support, so their tier can differ from the provider's posted-price tier.

NSECE community density measures the urban share of the nearby population, using tracts intersecting a two-mile circle around the anchor tract's population centroid and inverse-distance weights. High density means at least 85% urban, moderate means 30% to under 85%, and rural means under 30%.

Region	Child age	High density	Moderate	Rural
Northeast	0–1	6.73	6.40	5.09
	2–3	5.75	5.30	3.85
	4–5	5.33	4.52	3.58
Midwest	0–1	5.88	5.60	4.44
	2–3	4.75	4.38	3.18
	4–5	4.38	3.71	2.94
South	0–1	4.49	4.27	3.39
	2–3	3.94	3.63	2.64
	4–5	3.73	3.16	2.51
West	0–1	6.36	6.05	4.81
	2–3	5.29	4.88	3.55
	4–5	4.89	4.15	3.29

Cutoffs are in 2019 dollars per hour, rounded to cents.

3 Local care-hour targets

3.1 Children in regular care

1. **Estimate the state participation rate.** I pool the 2016–2019 NSCH observations for children ages 0–4 with a valid yes/no answer on at least ten weekly hours of non-parental care. The rate is the sum of child survey weights for yes answers divided by the sum for all valid answers. Pooling four pre-pandemic years reduces sampling noise in state estimates. NYC uses New York's rate because NSCH does not identify the city.

2. **Convert the rate to children using the model's care definition.** I multiply the rate by the Census 2019 population ages 0–4, adding county population estimates within each geography. The national factor 1.098054 is the 2019 NSECE child-weighted count of children ages 0–4 in regular non-parental care divided by the count with at least ten total weekly hours of that care, excluding school programs. It converts the NSCH ten-hour definition to NSECE's regular-care definition, which starts at five hours:

$$C_g^{\text{regular}} = P_{g,0\text{–}4}\, r_g^{\text{NSCH}}\,(1.098054).$$

   For the 50 states and DC, the result is 11.6 million children in regular care, 59.2% of 19.6 million children ages 0–4. These are population estimates, not unweighted survey counts.

3.2 Children in unpaid centers

1. **Pre-K.** I take the larger of public-school pre-K enrollment from the Common Core of Data and all-settings state pre-K enrollment from Table 2 of the 2019 NIEER yearbook. NYC uses its own all-settings city count. Taking the larger count includes public-school programs outside the state-funded program and state-funded programs outside public schools without adding the overlapping totals.

2. **Head Start and Early Head Start.** I convert funded slots to point-in-time enrolled children using national 2018–19 Office of Head Start enrollment-to-slot ratios: about 0.957 for Head Start and 0.923 for Early Head Start. For Early Head Start, I first remove the

3.592% share serving pregnant women. I remove home-visiting and family-child-care options using national program shares of about 1.94% for Head Start and 38.0% for Early Head Start.

3. **Remove children outside the modeled ages.** National 2019 NHES child-weighted shares, with age measured on December 31, remove 24.40% of pre-K children and 10.74% of Head Start children as age five or older. Early Head Start needs no age-five adjustment. NYC's pre-K calendar-year eligibility already keeps its count within ages 0–4. For New York State, I apply the national pre-K share only to the enrollment outside NYC.

4. **Remove pre-K/Head Start overlap.** I calculate one pooled overlap share o: the Head-Start-slot-weighted mean of the overlap shares in the 18 state/DC geographies with usable NIEER all-settings and unduplicated counts. I apply this pooled share, $o = 27.52\%$, to adjusted Head Start in every geography:

$$U_g = \text{adjusted pre-K}_g + (1 - o)\text{adjusted Head Start}_g + \text{adjusted Early Head Start}_g.$$

The pooled share supplies an overlap estimate where state counts alone cannot identify the same child in both programs. U_g can still overstate the model's unpaid-center category when programs charge fees or children also attend paid care.

3.3 Allocate children and convert to annual hours

1. **Predict the local care mix.** I apply the national model to the geography's ACS families. For each care type t, I sum each child's probability of that type times the child weight to obtain $B_{g,t}$. I exclude parent-only care and combine the two paid-center tiers. These are predicted population counts of children.

2. **Allocate the remaining children.** After setting unpaid-center children to U_g, I distribute $C_g^{\text{regular}} - U_g$ across paid center, paid home, unpaid home, other paid, and other unpaid care in proportion to their B values:

$$C_{g,t} = (C_g^{\text{regular}} - U_g)\frac{B_{g,t}}{\sum_{k\in\text{five types}} B_{g,k}}.$$

The survey participation rate and program counts set the level of care; the national model's predictions for local families supply the detail across types.

3. **Calculate six annual-hour targets.** For each type, including unpaid centers, I use the predicted part-time share $q_{g,t} = B_{g,t,\text{PT}}/(B_{g,t,\text{PT}} + B_{g,t,\text{FT}})$. At 10 weekly hours for part-time care or 40 for full-time care, for 50 weeks:

$$H_{g,t} = C_{g,t}\,[500q_{g,t} + 2{,}000(1 - q_{g,t})].$$

Using the model's hour grid makes these targets comparable with simulated care hours; the targets are annual child-hours.

3.4 Fit local baseline preferences

1. **Express the national preferences at local 2019 prices.** Let V be the financial utility of a choice and let choice 1 be the reference choice. For family i and choice j, I set

$$\alpha'_{ij} = \alpha_{ij} + (V_{ij}^{\text{national}} - V_{ij}^{\text{local}}) - (V_{i1}^{\text{national}} - V_{i1}^{\text{local}}).$$

This offsets the initial financial effect of substituting local prices and preserves the national model's predicted choice probabilities, p_0, before I fit the local targets.

2. **Match care hours and employment together.** I add one common preference shift δ_t per care type to every child's choices in the geography, summing the shifts for choices involving multiple children. Parent-only care has shift zero; the two paid-center tiers share one shift. I also add a working-choice shift for each primary-caregiver group: male/female by under 25, 25–34, 35–44, and 45+. I solve for these shifts jointly so that child-weighted care hours equal the six H targets and person-weighted employment rates equal the groups' mean employment-forest predictions. Joint fitting keeps both sets of targets satisfied.

3. **Use the fitted baseline for policy comparisons.** I hold these preferences fixed when policy changes. Because I fit hours, child counts and part-time shares need not equal those used to construct H. The model determines the split between low- and high-price centers.

4 Local supply inputs

4.1 Care-worker wages

1. **Fix the occupation mix by worker education.** I use national 2015–2019 ACS workers living in households, with positive earnings and usual work hours. Childcare workers exclude private-household employment; teachers and administrators must work in child day-care services. I calculate occupation shares using person weights, separately for workers without and with a bachelor's degree. Preschool special-education teachers enter the preschool-teacher wage category. The resulting childcare-worker, preschool-teacher, and administrator weights are approximately 72.6%, 24.2%, 3.2% without a BA and 42.6%, 45.8%, 11.6% with a BA.

2. **Calculate a local wage index.** I take the weighted mean of BLS OEWS median hourly wages for childcare workers (39-9011), preschool teachers (25-2011), and preschool/daycare administrators (11-9031), using the fixed ACS weights. For geography g, education group e, and year y, let $R_{g,e,y}$ be this local mean divided by the corresponding national mean. I multiply the national 2019 model wage by $R_{g,e,2019}$:

$$w_{g,e,2019} = w_{\mathrm{US},e,2019} R_{g,e,2019}, \qquad w_{\mathrm{US},e,2019} = \begin{cases} 12.06 & \text{without BA,} \\ 19.70 & \text{with BA.} \end{cases}$$

   Fixed national occupation weights keep differences in the occupation mix from driving the geographic wage index. NYC uses the New York metro area, including its New Jersey and Pennsylvania parts.

3. **Project wages.** I multiply the local 2019 wage by national nominal hourly-wage growth and by $R_{g,e,y}/R_{g,e,2019}$. The relative wage paths use 2019–2025 BLS data. Within each geography and education group, an occupation suppressed in any year is omitted from both local and national averages throughout the annual path, with the remaining weights renormalized. After 2025 I hold the relative ratio fixed and continue national wage growth. BLS's 2021 methodology change limits comparisons across years.

4.2 Staff hours per child-hour

1. **Select the rule in force on 1 October 2019.** For centers, I use children per adult at 11 months, 18 months, 35 months, 3 years, and 4 years, counting required teachers and aides. These five points represent the full years of age 0, 1, 2, 3, and 4; cutoffs within a year are approximated. For homes, I use regulated capacity excluding school-age-only additions. NYC

centers use the city Health Code; the other geographies use state rules. These legal limits measure differences in required staffing, not providers' observed staffing choices.

2. **Average the center requirements over ages.** I take the inverse of each children-per-adult ratio and average using national 2019 NSECE weighted center enrollment by age. The age 0 through 4 enrollment weights are 426,000, 718,000, 1,010,000, 2,170,000, and 2,340,000 children. With $c_{g,a}$ children per adult and n_a national enrollment,

$$S_{g,\text{center}} = \frac{\sum_{a=0}^{4} n_a / c_{g,a}}{\sum_{a=0}^{4} n_a}, \qquad S_{g,\text{home}} = 1/\text{selected home capacity}_g.$$

Common age weights make the index reflect rules rather than differences in the ages of children served across states.

3. **Scale the national labor requirements.** I divide each S by its average across the 50 states and DC, weighted by Census 2019 population ages 0–4. NYC is excluded from that national average. I multiply national staff hours per child-hour by this index, for both education groups. The center index applies to low-price, high-price, and unpaid centers; the home index applies to paid homes. An index of 1.2 raises the labor requirement by 20%.

4. **Keep the scope of home capacity explicit.** Selected home capacity is a rough staffing measure. Arkansas's 16, Rhode Island's 8, and Washington's 12 require additional staff; taking inverse capacity does not fully capture those assistant costs. Arizona uses four certified paid places and Louisiana uses six certified places. Age limits, own-child counting, and assistant requirements are recorded with each selected input. I do not separately weight municipal rules within a state, apart from the separate NYC run.

4.3 Non-labor costs

1. **Start with each national residual.** The amount left after labor costs in the national 2019 price is \$0.510055 per child-hour for low-price centers, \$4.614567 for high-price centers, and \$2.827343 for paid homes. This residual includes non-wage costs and provider margins. Unpaid centers have residual zero.

2. **Apply a fixed geographic cost factor and annual inflation.** I multiply each residual by the geography's 2019 BEA housing-services RPP divided by 100, then by CPI growth since 2019:

$$d_{g,t,y} = d_{\text{US},t,2019} \frac{\text{housing RPP}_{g,2019}}{100} \frac{\text{CPI}_y}{\text{CPI}_{2019}}.$$

Housing services provide a proxy for geographic differences in premises and other non-labor costs. CPI increases the nominal residual over time, consistently with the national model. NYC uses the New York metro RPP. The geographic factor stays at its 2019 value.

4.4 Size the care-worker labor market

At local 2019 target care hours, I calculate each education group's required staff hours by multiplying sector care hours by the local staff-hours-per-child-hour requirement and summing across sectors. This is $L_{0,g,e}$. For employment L, the worker wage is

$$w_{g,e} = w_{g,e}^{\text{base}} \left(\frac{L}{L_{0,g,e}} \right)^{1/\varepsilon_e}, \qquad \varepsilon_{\text{no BA}} = 4, \quad \varepsilon_{\text{BA}} = 2.$$

Thus, baseline staff hours command the local base wage; increased care demand raises wages according to the national labor-supply elasticities.

5 Parents' employment paths

1. **Estimate relative local employment rates.** For each sex and fine age cell, I pool two adjacent ACS years and divide weighted employed people by weighted people in that cell, using person weights. I divide the local rate by the national ACS rate from the same two-year pool. Pooling reduces noise in small local age cells. The fine ages are 16–17, 18–19, 20–24, and five-year bins from 25–29 through 55–59.

2. **Apply the national projection.** I multiply the corresponding national CPS/CBO projected employment rate by this local-to-national ratio. Observed relative ratios run through the 2023–2024 pool; I hold that last ratio fixed afterward. This combines national forecast changes with each place's latest relative employment position.

3. **Translate the path to caregiver targets.** I combine fine age cells into the model's under-25, 25–34, 35–44, and 45+ groups, separately by sex, using the geography's fixed 2019 ACS age-population weights. The 45+ path uses the available 45–59 projection cells. I multiply each group's matched 2019 employment-forest rate by growth in its projected local rate since 2019, capping the target below one. I fit working-choice shifts to these targets in each baseline year and hold them fixed for that year's policy comparisons.

6 Implementation and checks

1. **Inputs built.** Demand and supply inputs cover all 52 geographies. The staffing review is complete: 312 unweighted geography/setting/age records, comprising five center ages and one home input per geography. Of these, 28 change a numerical input (15 center age points and 13 home inputs); 284 confirm the prior value. All are incorporated in the supply build; historical-source and hash checks pass. All 52 annual wage paths match their source data.

2. **Baseline fitting.** The demand check recovers p_0 after the initial price substitution and matches care-hour targets to relative gaps below 10^{-8}. NY minus NYC and a separately constructed rest-of-state target have equal unpaid-center child counts but different part-time shares. Those shares explain the 1.04 million annual-child-hour difference (1.22% of unpaid-center hours, 0.163% of all rest-of-state care hours). The strict hours-additivity diagnostic still fails for this reason.

3. **Price validation.** I compare model full-time weekly prices with NDCP county median prices, including NDCP's filled-in values. I aggregate county prices using Census population ages 0–4 and model prices using local predicted paid-center enrollment across tiers and ages. Model hourly prices become 40-hour weeks. I match model ages 0–1, 2, and 3–4 to NDCP's infant (0–23 months), toddler (24–35), and preschool (36–54) groups. Model four-year-olds also include months 55–59, and NDCP survey definitions of full-time care vary by state. The comparison checks price levels and geographic patterns with these measurement limits.

   With the updated staffing inputs, the 2019 comparison covers 46 states: the unweighted correlation is 0.892 and weekly RMSE is \$34.73. The child-weighted mean model price is 11.8% above the NDCP mean. Weekly RMSE is \$39.20, \$41.71, and \$47.35 in 2020–2022, covering 46, 41, and 45 states respectively.

4. **Simulation validation.** The check uses a 1% ACS sample for 2026–2027 across all 52 geographies: 104 geography-year solves. A solve must converge, have a paid-sector price gap below $0.01 per child-hour, and miss each employment target by less than 0.001 percentage points. The 22 October run with all current numerical staffing corrections passed all 104 solves. The largest paid-sector price gap was $0.000319 per child-hour and the largest employment miss was 0.0002545 percentage points.

6.1 Open items

No unfinished staffing, wage, price-comparison, or baseline-simulation tasks remain from this update. The NY/NYC hours diagnostic remains flagged as described above.

`SOURCES.csv` lists source links and dates. The staffing evidence register records the rule, effective date, selected value, and conditions.